10. Diagnostic Procedure with Diagnostic Trouble Code (DTC)

A: DTC P1571 REFERENCE CODE INCOMPATIBILITY

DTC DETECTING CONDITION:

Reference code incompatibility between IMMCM and ECM

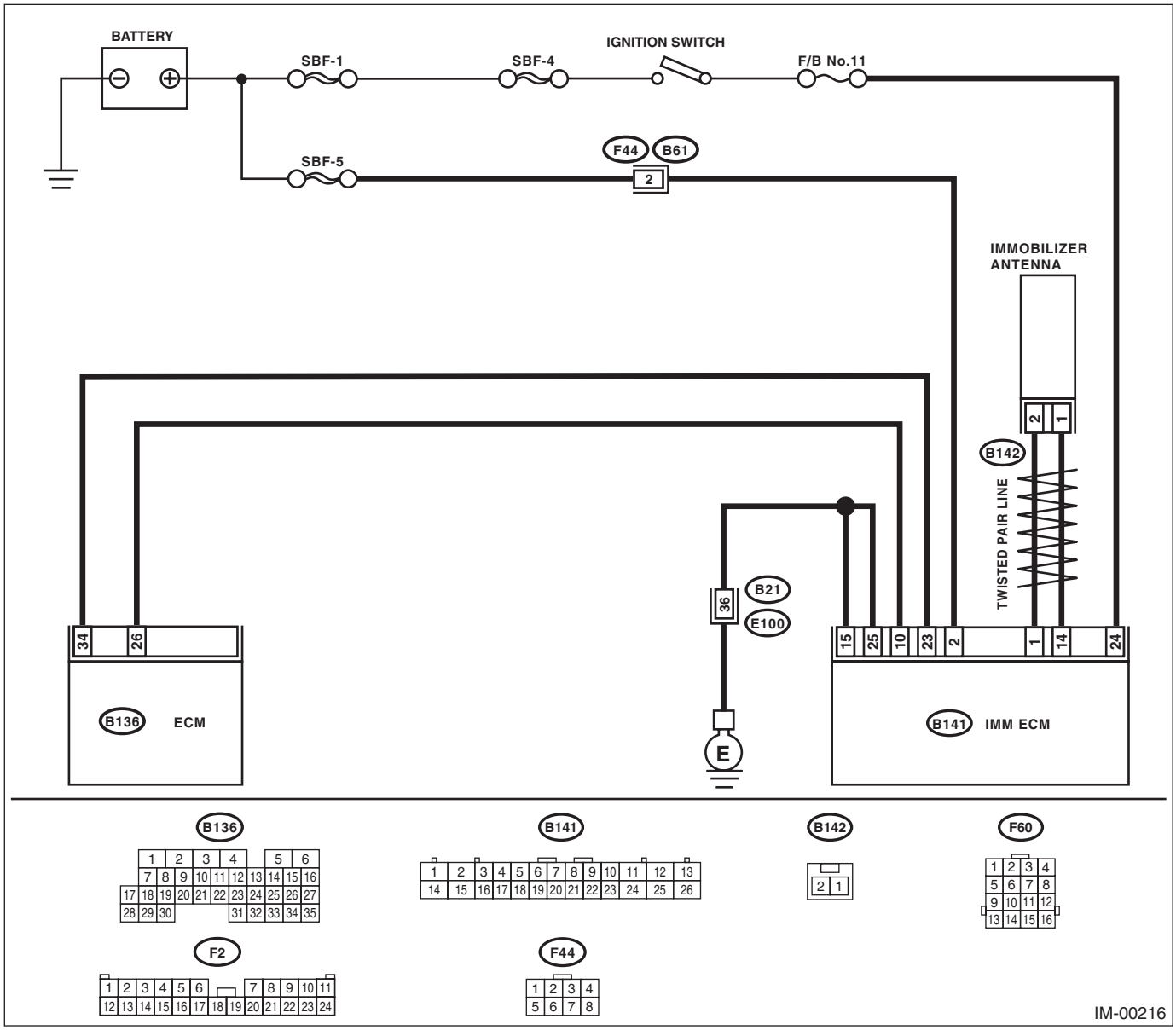
Step		Check	Yes	No
1	**PERFORM REGISTRATION ON IGNITION KEY.** Perform registration on all keys of the vehicle. Refer to the “REGISTRATION MANUAL FOR IMMOBILIZER”.	Is registration for all keys complete?	Finish the diagnosis.	Go to step **2**.
2	**CHECK DTC.**	Is there any immobilizer-related DTC except DTC P1571?	Eliminate the cause of DTC other than DTC P1571, and perform registration on keys again.	Replace the ECM and IMMCM <Ref. to FU(H4DOTC)-50, Engine Control Module (ECM).> <Ref. to SL-47, Immobilizer Control Module.>, and then replace all ignition keys (including the transponder). Execute the registration procedure next. Refer to the “REGISTRATION MANUAL FOR IMMOBILIZER”.

B: DTC P1572 IMM CIRCUIT FAILURE (EXCEPT ANTENNA CIRCUIT)

DTC DETECTING CONDITION:

Communication failure between IMMCM and ECM

WIRING DIAGRAM:



Step		Check	Yes	No
1	**CHECK IMMCM POWER SUPPLY CIRCUIT.** 1) Turn the ignition switch to OFF. 2) Disconnect the harness connector from IMMCM. 3) Measure the voltage between IMMCM harness connector terminal and chassis ground. ***Connector & terminal*** ***(B141) No. 2 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **2**.	Check the harness for open or short between IMMCM and fuse.
2	**CHECK IGNITION SWITCH CIRCUIT.** 1) Turn the ignition switch to ON. (engine OFF) 2) Measure the voltage between IMMCM harness connector terminal and chassis ground. ***Connector & terminal*** ***(B141) No. 24 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **3**.	Check the harness for open or short between IMMCM and ignition switch.

	Step	Check	Yes	No
3	**CHECK IMMCM GROUND CIRCUIT.** 1) Turn the ignition switch to OFF. 2) Measure the resistance between IMMCM harness connector terminal and chassis ground. ***Connector & terminal*** ***(B141) No. 15 — Chassis ground:*** ***(B141) No. 25 — Chassis ground:***	Is the resistance less than 10 Ω?	Go to step **4**.	Repair the open circuit of IMMCM ground circuit.
4	**CHECK HARNESS BETWEEN IMMCM AND ECM.** 1) Disconnect the harness connector from the ECM and IMMCM. 2) Measure the resistance between IMMCM harness connector terminal and ECM harness connector terminal. ***Connector & terminal*** ***(B141) No. 10 — (B136) No. 26:***	Is the resistance less than 10 Ω?	Go to step **5**.	Repair the open circuit of harness between IMMCM and ECM.
5	**CHECK HARNESS BETWEEN IMMCM AND ECM.** Measure the resistance between IMMCM harness connector terminal and ECM harness connector terminal. ***Connector & terminal*** ***(B141) No. 23 — (B136) No. 34:***	Is the resistance less than 10 Ω?	Go to step **6**.	Repair the open circuit of harness between IMMCM and ECM.
6	**CHECK HARNESS OF COMMUNICATION LINE.** 1) Turn the ignition switch to ON. (engine OFF) 2) Measure the voltage between IMMCM harness connector terminal and chassis ground. ***Connector & terminal*** ***(B141) No. 10 (+) — Chassis ground (–):*** ***(B141) No. 23 (+) — Chassis ground (–):***	Is the voltage less than 1 V?	Go to step **7**.	There is a short circuit in the battery voltage circuit or ignition switch "ON" circuit. Repair the harness between IMMCM and ECM.
7	**CHECK HARNESS OF COMMUNICATION LINE.** Measure voltage between the ECM harness connector terminal and engine ground. ***Connector & terminal*** ***(B136) No. 26 (+) — Engine ground (–):*** ***(B136) No. 34 (+) — Engine ground (–):***	Is the voltage less than 1 V?	Go to step **8**.	There is a short circuit in the battery voltage circuit or ignition switch "ON" circuit. Repair the harness between IMMCM and ECM.
8	**CHECK ECM BY COMMUNICATION LINE CHECK.** 1) Connect the harness connector to ECM. 2) Disconnect the harness connector from IMMCM. 3) Perform communication line check. <Ref. to IM(diag)-7, COMMUNICATION LINE CHECK, OPERATION, Subaru Select Monitor.>	Does "Communication Line not Shorted" appear on the screen?	Replace the IMMCM <Ref. to SL-47, Immobilizer Control Module.> and then replace all ignition keys (including the transponder). Execute the registration procedure next. Refer to the "REGISTRATION MANUAL FOR IMMOBILIZER".	Replace the ECM. <Ref. to FU(H4DOTC)-50, Engine Control Module (ECM).> Then perform registration operation. Refer to the "REGISTRATION MANUAL FOR IMMOBILIZER".

C: DTC P1574 KEY COMMUNICATION FAILURE

DTC DETECTING CONDITION:

- Failure of IMMCM to verify key (transponder) ID code
- Key (transponder) malfunctioning

Step		Check	Yes	No
1	**CHECK IMMCM FUNCTION.** Insert the key into the ignition switch (LOCK position), then measure changes in voltage between the antenna connectors. ***Connector & terminal*** ***(B142) No. 1 (+) — No. 2 (–):***	After inserting the key, is the voltage 0 — 30 V (Approx. 0.1 second) or 0 V (Approx. 1 second)?	Go to step **2**.	Replace the IMMCM <Ref. to SL-47, Immobilizer Control Module.> and then replace all ignition keys (including the transponder). Execute the registration procedure next. Refer to the "REGISTRATION MANUAL FOR IMMOBILIZER".
2	**CHECK IGNITION KEY (TRANSPONDER).** 1) Remove the key from ignition switch. 2) Start the engine using other key which is already registered.	Does the engine start?	Replace the ignition key (including the transponder). Execute the registration procedure next. Refer to the "REGISTRATION MANUAL FOR IMMOBILIZER".	Replace the IMMCM <Ref. to SL-47, Immobilizer Control Module.> and then replace all ignition keys (including the transponder). Execute the registration procedure next. Refer to the "REGISTRATION MANUAL FOR IMMOBILIZER".

D: DTC P0513 INCORRECT IMMOBILIZER KEY

DTC DETECTING CONDITION:
Incorrect immobilizer key (Use of unregistered key in IMMCM)

Step		Check	Yes	No
1	**PERFORM REGISTRATION ON IGNITION KEY.** Perform registration on all keys of the vehicle. Refer to the “REGISTRATION MANUAL FOR IMMOBILIZER”.	Is registration for all keys complete?	Finish the diagnosis.	Replace all ignition keys (including the transponder). Go to step **2**.
2	**PERFORM REGISTRATION ON IGNITION KEY.** Perform registration on all keys of the vehicle. Refer to the “REGISTRATION MANUAL FOR IMMOBILIZER”.	Is registration for all keys complete?	Finish the diagnosis.	Replace the IMMCM <Ref. to SL-47, Immobilizer Control Module.> and then replace all ignition keys (including the transponder). Execute the registration procedure next. Refer to the “REGISTRATION MANUAL FOR IMMOBILIZER”.

E: DTC P1576 EGI CONTROL MODULE EEPROM

DTC DETECTING CONDITION:
ECM malfunctioning
ECM replacement
Replace the ECM.
<Ref. to FU(H4DOTC)-50, Engine Control Module (ECM).>
Then perform registration operation. Refer to the “REGISTRATION MANUAL FOR IMMOBILIZER”.

F: DTC P1577 IMM CONTROL MODULE EEPROM

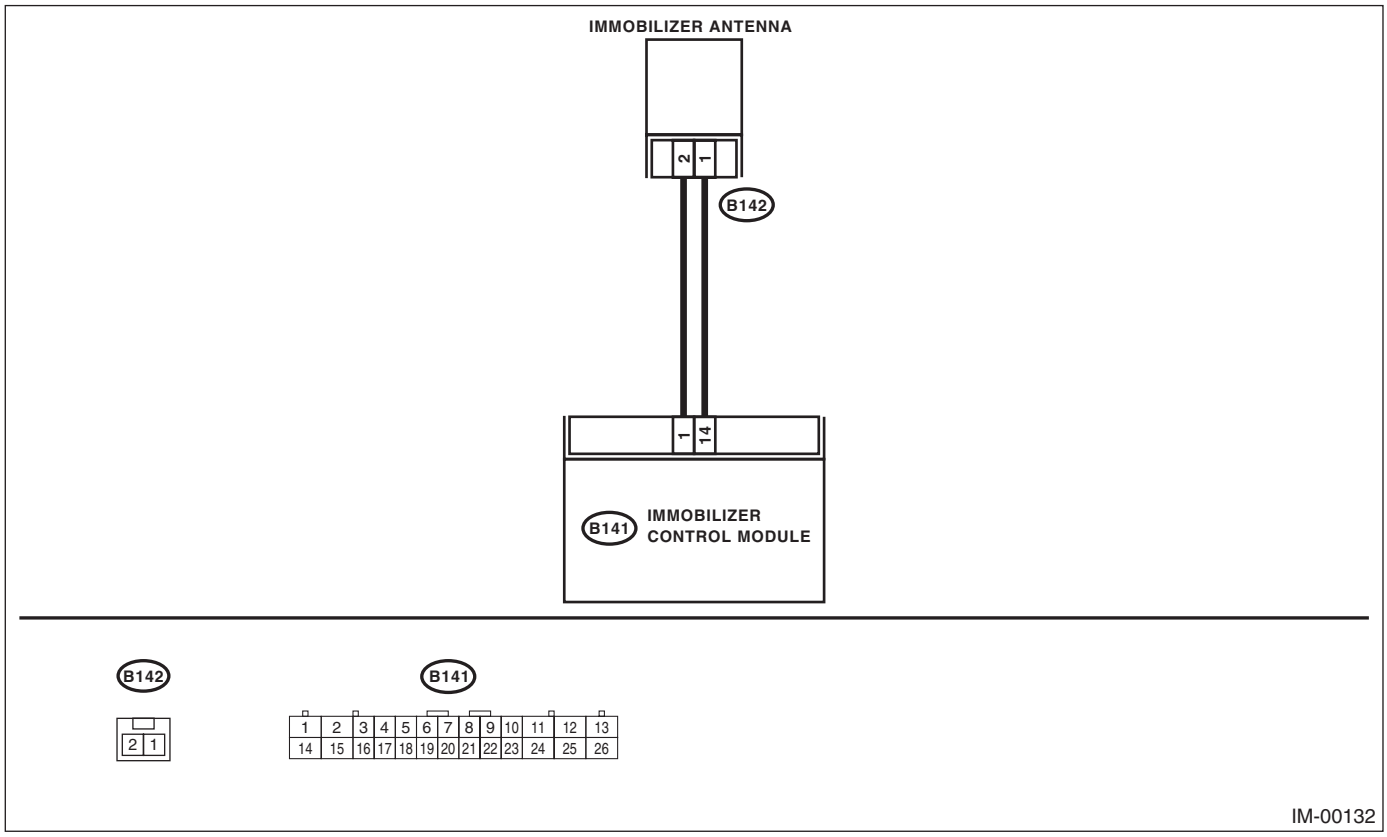
DTC DETECTING CONDITION:
IMMCM malfunctioning
IMMCM replacement
Replace the IMMCM <Ref. to SL-47, Immobilizer Control Module.> and then replace all ignition keys (including the transponder). Execute the registration procedure next. Refer to the “REGISTRATION MANUAL FOR IMMOBILIZER”.

G: DTC P1570 ANTENNA

DTC DETECTING CONDITION:

Faulty antenna

WIRING DIAGRAM:



Step		Check	Yes	No
1	**CHECK ANTENNA CIRCUIT.** 1) Turn the ignition switch to OFF. 2) Disconnect the harness antenna connector from IMMCM. <Ref. to SL-48, Immobilizer Antenna.> 3) Measure the resistance of the antenna circuit. ***Connector & terminal*** ***(B142) No. 1 — No. 2:***	Is the resistance less than 10 Ω?	Go to step **2**.	Replace the antenna. <Ref. to SL-48, Immobilizer Antenna.>
2	**CHECK ANTENNA CIRCUIT.** Measure the resistance between antenna harness connector and chassis ground. ***Connector & terminal*** ***(B141) No. 1 — Chassis ground:***	Is the resistance less than 10 Ω?	Replace the antenna. <Ref. to SL-48, Immobilizer Antenna.>	Go to step **3**.
3	**CHECK ANTENNA CIRCUIT.** Measure the resistance between antenna harness connector and chassis ground. ***Connector & terminal*** ***(B141) No. 14 — Chassis ground:***	Is the resistance less than 10 Ω?	Replace the antenna. <Ref. to SL-48, Immobilizer Antenna.>	Go to step **4**.
4	**CHECK ANTENNA CIRCUIT.** 1) Turn the ignition switch to ON. (engine OFF) 2) Measure the voltage between antenna harness connector and chassis ground. ***Connector & terminal*** ***(B141) No. 1 (+) — Chassis ground (–):***	Is the voltage less than 1 V?	Go to step **5**.	Replace the antenna. <Ref. to SL-48, Immobilizer Antenna.>

Step		Check	Yes	No
5	**CHECK ANTENNA CIRCUIT.** Measure the voltage between antenna harness connector and chassis ground. ***Connector & terminal*** ***(B141) No. 14 (+) — Chassis ground (–):***	Is the voltage less than 1 V?	Go to step **6**.	Replace the antenna. <Ref. to SL-48, Immobilizer Antenna.>
6	**CHECK IMMCM FUNCTION.** 1) Turn the ignition switch to OFF. 2) Connect the antenna harness connector to IMMCM. 3) Insert the key into the ignition switch, then measure changes in voltage between the antenna harness connectors. ***Connector & terminal*** ***(B141) No. 1 (+) — No. 14 (–):***	After inserting the key, is the voltage 0 — 30 V (Approx. 0.1 second) or 0 V (Approx. 1 second)?	Go to step **7**.	Replace the IMMCM <Ref. to SL-47, Immobilizer Control Module.> and then replace all ignition keys (including the transponder). Execute the registration procedure next. Refer to the "REGISTRATION MANUAL FOR IMMOBILIZER".
7	**CHECK IGNITION KEY (TRANSPONDER).** 1) Remove the key from ignition switch. 2) Start the engine using other key which is already registered.	Does the engine start?	Replace the ignition key (including the transponder). Execute the registration procedure next. Refer to the "REGISTRATION MANUAL FOR IMMOBILIZER".	Replace the IMMCM <Ref. to SL-47, Immobilizer Control Module.> and then replace all ignition keys (including the transponder). Execute the registration procedure next. Refer to the "REGISTRATION MANUAL FOR IMMOBILIZER".

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